

5.3.1 The Fundamental Theorem of Calculus

Introduction: The first part of the fundamental theorem of calculus deals with functions of the form:

$$g(x) = \int_a^x f(t) dt .$$

Question to the Class: Do you see any *awesome* relationship between the function of integration in the introduction and the formula we obtained for $g(x)$?

Theorem (The Fundamental Theorem of Calculus, Part 1): If f is a continuous function on $[a, b]$, then the function g defined by

$$g(x) = \int_a^x f(t) dt \text{ where } a \leq x \leq b$$

Proof:

Let f be a continuous function on $[a, b]$ and let $g(x) = \int_a^x f(t) dt$ where $a \leq x \leq b$.

We will try to determine the value of $g'(x)$.

Example 1: Find the first derivative of the function $g(x) = \int_0^x \sqrt{1+t^2} dt$.

Example 2: Find the first derivative of the function $g(x) = \int_x^{10} (t - e^t) dt$.

Example 3: Find the first derivative of the function $k(x) = \int_1^{x^4} \frac{\sin(t)}{t} dt$.

Example 4: Find the first derivative of the function $k(x) = \int_3^{\sqrt{x}} \frac{z^2}{z^4 + 1} dz$

While we see how the Fundamental Theorem of Calculus (part 1) can be used, we may still question why it is significant. What relevance does this idea have to calculating definite integrals (finding areas under curves)?

Let's consider the following implication of the FTC1.

Let $g(x) = \int_a^x f(t) dt$. We know from the FTC1 that if $x \in (a, b)$ and $f(t)$ is continuous on $[a, b]$ then $g'(x) = f(x)$.

Theorem (Fundamental Theorem of Calculus, Part 2): If f is continuous on $[a, b]$, then

Note: We will use the FTC2 A LOT in finding definite integrals! In fact, this is the theorem that will free us from having to evaluate very difficult sums!!!

Example 5: Evaluate $\int_{-1}^3 (x^2 - 2) dx$.

Example 6: In section 5.2.1 we saw that $\int_{-5}^1 (\sqrt{2-x}) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\sqrt{7 - \frac{6i}{n}} \right) \frac{6}{n}$ and we mentioned that the sum on the right is too difficult for us to figure out. Use the FTC2 to find $\int_{-5}^1 (\sqrt{2-x}) dx$.

Example 7: Find the exact value of $\int_{-1}^1 \frac{1}{1+x^2} dx$.

Example 8: Find the exact value of each of the following.

a) $\int_1^3 e^x dx$.

b) $\int_{-3}^{-6} \frac{dx}{x}$

c) The area under the cosine curve from 0 to $\frac{\pi}{4}$.

Example 9: Jeff attempted to compute the following integral but was told by his teacher that his answer is wrong. What is wrong with Jeff's calculation?

$$\int_{-1}^3 \frac{1}{x^2} dx = -\frac{1}{x} \Big|_{-1}^3 = \left(-\frac{1}{3}\right) - \left(-\frac{1}{(-1)}\right) = -\frac{1}{3} - 1 = -\frac{4}{3}$$